

SKILLRL: Evolving Agents via Recursive Skill-Augmented Reinforcement Learning

WebShop

Task: I need a women's long sleeve button-down shirt in navy blue, size large, machine washable, price lower than \$40.00

--- Retrieved Skills ---

Detected Task Type: apparel

[General Skills] ... [Task-Specific Skills]

Step 1

Observation: WebShop homepage. Search bar available.

<think>Following skill [gen_001] 'Prioritize Core Keywords' and [app_001] 'Focus Key Query': I should include product type + must-have attributes. My query should be ...</think>
<action>search[women long sleeve button down shirt navy blue machine ...]</action>

...

Step 9

Observation: Product page updated: Color = Navy Blue, Size = L. Price: \$29.99

<think>Applying skill [app_004] 'Check Variant Price': price for Navy Blue / L is \$29.99, which is under the \$40 budget. Also per [app_005] 'Confirm Hidden Specs': ... \$29.99 <\$40. Per skill [gen_013] 'Purchase Decisively', I should buy now.</think>
<action>click[Buy Now]</action> ✓

ALFWorld

Task: heat some egg and put it in countertop.

--- Retrieved Skills ---

Detected Task Type: heat

[General Skills] ... [Task-Specific Skills]

Step 1

Observation: You are in the middle of a room. Looking quickly around you...

<think>Per [gen_010] 'Progressive Goal Decomposition': sub-goals are ... Per [hea_004] 'No Appliance Before Object': I must NOT go to microwave first. Per [hea_001] 'Secure Exact Target First': find the egg. It's food — check fridge and countertops.</think>
<action>go to countertop 1</action>

...

Step 8

Observation: You arrive at countertop 1

<think>Per [hea_005]: place the heated egg directly.</think>
<action>put egg 1 in/on countertop 1</action> ✓

Figure 6. Case studies of SKILLRL on WebShop and ALFWorld. The examples illustrate how the agent adaptively retrieves and integrates General Skills and Task-Specific Skills within its reasoning process to achieve precise and efficient task execution.

2025). While RL is widely used to align LLMs (Schulman et al., 2017; Ouyang et al., 2022), or improve reasoning via rule-based verifiers (Shao et al., 2024), applying it to agentic skills remains challenging due to sparse rewards and long horizons. Unlike previous memory-augmented RL

2024.

Anthropic. The claude 3 model family: Opus, sonnet, haiku, 2024. URL <https://www.anthropic.com/news/claude-3-family>.